EXPRESSION OF INTEREST APPLICATION
2025-2026

Privacy Statement

The information collected in this application will be used, and may be disclosed, for the purposes of assessing the merits of the application. As part of the assessment process, the information may be shared with external consultants, review committee members, officials in other departments, federal, provincial and/or territorial governments or Members of Parliament. It may also be used and/or disclosed for policy analysis, research, and/or evaluation purposes. In order to conduct these activities, various sources of information under the custody and control of the Canadian Northern Economic Development Agency (CanNor) may be linked. However, these additional uses and/or disclosures of information will not impact the project. In the event that the application contains personal information, the personal information will be administered in accordance with the *Privacy Act* (https://laws-lois.justice.gc.ca/eng/acts/P-21/index.html). The application is also subject to the *Access to Information Act* (https://www.canada.ca/en/treasury-board-secretariat/services/access- information-privacy.html) provides every person with a right of access to information under the control of the department, subject to a limited set of exemptions. Instructions for obtaining access to this information are outlined in the government publication entitled *Info Source* (https://www.cannor.gc.ca/eng/1395919423905

Applicant and Contact Information	
Applicant legal name City of Iqaluit	Project name
Primary contact name Bill Williams	Telephone number 867.975.8521
Email address w.williams@iqaluit.ca	Applicant business number (CRA 9 digit identifier) ☐☐☐☐☐☐☐☐☐
Applicant mailing address	

Applicant type (select one option that applies)

- ☐ For-Profit business
- ☐ Non-Profit Association
- ☐ Indigenous Government
- ☐ Territorial government
- X Municipal Government
- ☐ Other (please specify):

Sector type (see EOI Guide Section 4.8 for more information)

Applicant Sector:
913 Local, municipal and regional public administration - City of Iqaluit

Describe your organization or business and its mandate:
The City of Iqaluit,(COI) is responsible for providing essential municipal services and infrastructure to support the well-being and growth of the community. Its mandate includes delivering essential infrastructure and services while promoting economic development and enhancing the quality of life for its residents.

Project information		
Project Location Address: (Community, Territory) Iqaluit, Nunavut	Expected start date (YYYYMMDD) 2025/06/01	Expected end date (YYYMMDD) 2028/03/31

Project Description: the focus of the project should be clear and demonstrate how an existing opportunity or challenge will be addressed. Please provide answers to the following questions. **Please refer to Section 4.14 of the EOI Guide before answering questions.**

- Overview:
- Describe the project.
- Outline key activities required to complete the project.
- Outline the proposed project timeline.

See EOI Guide for more details and examples.

Since October 2024, the City of Iqaluit began working on a series of impactful projects with University of Waterloo students to address critical municipal needs through innovative research and technology. These initial projects—focusing on software optimization for asset management, AI-driven predictive scheduling for water and sewer services, and AI-powered land use planning—are just the beginning of a long-term partnership aimed at tackling real-world infrastructure projects in Iqaluit utilizing large-language models. The City of Iqaluit aims to set a pioneering example for all of Nunavut by demonstrating how AI can be effectively leveraged to address municipal challenges, fostering innovation and resilience across the territory. Through these projects, Iqaluit strives to lead the way in AI-driven solutions for northern communities.

Our goal is to continue this collaboration in the future, building on the momentum and insights gained in these early stages. By working closely with students and researchers, we envision expanding these efforts and identifying new project areas that align with Iqaluit's evolving priorities. This sustained partnership will not only benefit the City but will also offer students invaluable hands-on experience in developing AI/ML and LLM solutions for specifically for northern, remote, and uniquely challenging environments. Through continuous engagement, we hope to drive lasting change and further Iqaluit’s goals for sustainable growth and efficient city management.

Outlined below are descriptions of the AI-focused projects currently underway with the University of Waterloo. To expand the impact of these initiatives and support future projects, we are seeking funding from CanNor. This support would enable us to drive these developments further, ensuring sustained progress and innovation in our work.

Software Stack Analysis and Recommendation for Municipal Asset Management and Works Tasking
This project aims to evaluate and improve Iqaluit's current software solutions for municipal asset management. Key objectives include analyzing existing tools, identifying system inefficiencies, and recommending an integrated software stack tailored to the city’s needs. The scope includes reviewing best practices, conducting an environmental scan, and developing a prototype solution. Expected deliverables are a comprehensive report on software gaps, a prototype stack or architecture plan, and a presentation summarizing findings and recommendations.

AI/ML-Powered Trucked Water and Sewer Service Predictive Scheduling
Focused on enhancing the efficiency of Iqaluit's trucked water and sewer services, this project seeks to minimize fuel usage and improve service reliability. Key objectives include developing predictive AI/ML models for household water consumption and optimizing delivery routes in real-time. The work involves data collection, algorithm development, and the creation of an app-based route guide. Deliverables include a predictive model, an optimized scheduling algorithm, a routing app, and a proof-of-concept demonstration with a technical report.

AI/ML-Powered Land Use Planning
This project addresses the need for optimized land use planning in Iqaluit by incorporating AI/ML to support sustainable urban growth. Objectives include creating AI-driven tools that analyze topography, geotechnical data, zoning laws, and community needs to develop optimized land use plans. The project entails data gathering, algorithm development, and generating actionable land use proposals. Deliverables include an AI/ML-powered land use planning tool, detailed land use plans with supporting analysis, and a presentation on methodology and results.

Timelines

Start of Project

City of Iqaluit Sprint: The Sprint will be a focused, multi-day workshop designed to identify and address the City of Iqaluit's pressing needs in housing, mobility, sustainable infrastructure, integrated future city modeling, and AI/ML applications within municipal operations. This collaborative session will bring together city representatives, University of Waterloo students, researchers, and industry experts to analyze existing challenges, pinpoint key gaps, and co-develop innovative solutions.

This session will explore strategies for resilient infrastructure, cold-climate mobility, and sustainable growth, all tailored to Iqaluit's unique context. Participants will also examine how AI/ML can enhance delivery of municipal services, creating a roadmap for technology integration in areas like predictive maintenance and resource management. By the end of the Sprint, actionable recommendations and project outlines will be established, paving the way for impactful, community-centered advancements in Iqaluit's municipal operations.

Key Deliverables of the City of Iqaluit Municipal Sprint:

- **Comprehensive Needs Assessment:** A detailed report capturing the identified needs in core areas such as housing, mobility, and sustainable infrastructure, tailored to the unique challenges of Iqaluit.

- **Strategic Action Roadmap:** A prioritized plan with specific, scalable actions for each focus areas.

- **AI/ML Integration Strategy:** A targeted plan for implementing AI/ML tools within municipal operations, enhancing efficiency, data-driven decision-making, and service delivery.

Year 1 Initiatives

1. Capstone Collaboration with the City of Iqaluit: Building on insights from the Municipal Sprint, the City of Iqaluit will collaborate with the University of Waterloo's Capstone program to harness UW students' skills in advancing key projects. Over a 4-8 month period, 1-4 teams of 4-5 students will work closely with the City, co-developing innovative solutions through hands-on work in data analytics, AI integration, and prototyping. This partnership will foster impactful change for Nunavut communities.

2. Knowledge Exchange Workshop: This dynamic workshop will unite representatives from the City of Iqaluit with UW students and researchers for collaborative brainstorming and project planning. City representatives will outline pressing community challenges and share insights into local priorities and cultural contexts. Students will present their initial ideas and approaches, receiving direct feedback to refine their concepts. This iterative, two-way dialogue will help shape projects that are both innovative and deeply rooted in Iqaluit's unique needs, laying the groundwork for meaningful, community-driven outcomes.

3. AI/ML Research Network Engagement: The City of Iqaluit's project proposals will be shared in a competitive call across a network of over 200 AI-focused researchers. Within 2-3 weeks, the University of Waterloo will collect submissions from principal investigators, each detailing research methodologies, timelines, budgets, and graduate student support requirements. The Future Cities Institute will fund selected graduate and post-doctoral students to work on these projects, driving forward Iqaluit-directed initiatives with cutting-edge research.

Year 2 Initiatives

1. **Ongoing Research and Capstone Projects:** Building on the successes of Year 1, Year 2 will focus on advancing AI-driven tools, resources, and data tailored specifically for the City of Iqaluit and with broader applicability to the entire North. By continuously integrating feedback from city representatives, these tools will undergo iterative refinements, ensuring they remain relevant, effective, and closely aligned with the priorities and evolving needs of the Iqaluit community. This ongoing development will deepen the partnership's impact and create more robust solutions that address local challenges.

2. **Identification of New UW Capstone Projects:** Leveraging insights from the Municipal Sprint and priority feedback from the City of Iqaluit, new project opportunities will be identified for University of Waterloo students to explore. These projects will reflect the unique goals of the City and give students meaningful opportunities to apply their knowledge in real-world settings, driving forward practical innovations that serve Iqaluit's needs.

3. **City of Iqaluit Problem-Statement Hackathon:** In an intensive, two-week hackathon, over 100 University of Waterloo students from diverse disciplines will collaborate in teams to address 1-2 critical problem areas defined by the City of Iqaluit. This event will foster innovative thinking and allow students to develop creative solutions, providing Iqaluit with a range of actionable ideas to tackle local challenges. The hackathon will also foster skill-

building and creative problem-solving among participants, adding value to both the City and the students involved.

4. **AI Training and Continuing Education Program:** To support the integration and effective use of the developed AI tools, the University of Waterloo will offer a specialized 2-3 session course for representatives of the City of Iqaluit. Led by leading AI instructors, this program will provide hands-on, practical training in using AI applications tailored to community-focused projects. Participants will gain technical expertise, best practices, and confidence to integrate these tools into their work, enabling impactful, informed decision-making that supports community-driven outcomes.

5. **Municipal Sprint 2.0:** With a strong foundation of established research projects, this second Municipal Sprint will introduce new problem statements that aim to foster sustainable economic growth, enhance community capacity, and support long-term well-being in Iqaluit and the northern territories. This sprint will serve as a launchpad for fresh ideas, collaborative opportunities, and research-backed solutions, all centered on building resilience and sustainability for northern communities.

Potential projects emerging from Municipal Sprint 2.0:

The Municipal Sprint 2.0 will serve as a catalyst for identifying impactful, community-focused projects that address Iqaluit's unique challenges and priorities. Key areas of potential project development include, but are not limited to:

- Workforce Development: Projects to enhance the availability and quality of job skills training in AI/ML that align with current industry demands. These initiatives will help build a skilled workforce, equipped to meet the growing needs of both the city and its evolving job market.

- Food Security Innovations: Initiatives aimed at bolstering food sovereignty and security in Iqaluit through innovative food production methods, such as hydroponics and greenhouses, as well as the development of regional food supply chains. These projects will explore sustainable and resilient approaches to food production that reduce dependency on external sources.

- Sustainable Construction Methods: Projects focused on implementing and scaling advanced construction techniques, including modular or net-zero housing, 3D printing, and the development of sustainable building materials. Additionally, these projects may incorporate automation and efficiency improvements in residential construction to address housing shortages and reduce costs.

Through Municipal Sprint 2.0, these potential projects will be mapped out in detail, creating a structured roadmap for actionable, community-driven solutions that support long-term resilience and growth.

6. **AI/ML Research Network Engagement:** Following Municipal Sprint 2.0, identified projects from the City of Iqaluit will be shared within a network of over 200 AI-focused researchers in a competitive call for proposals. This process will attract top research talent, with FCI funding provided to graduate and post-doctoral students working on these initiatives, further advancing project goals and strengthening AI-driven solutions for the City.

Year 3 Initiatives

By the end of Year 3, all research projects initiated through the partnership between the City of Iqaluit and the University of Waterloo will reach completion, resulting in a comprehensive suite of practical solutions tailored to Iqaluit's unique needs and serving as a model for other communities across Nunavut. Each project will produce targeted, actionable outcomes—such as refined tools, data-driven reports, and validated prototypes—that support the City's sustainable development goals and demonstrate the potential of AI-driven municipal solutions throughout the North.

Key outcomes include:

- **Finalized AI Tools and Solutions:** The collaboration will deliver a suite of advanced AI-driven tools, such as optimized municipal asset management software, predictive scheduling systems for essential services, and enhanced land use planning models. These tools will empower the City of Iqaluit with cutting-edge capabilities for data-informed decision-making, operational efficiency, and sustainable urban management that can also be utilized in other Hamlets.

- Implementation Roadmaps: Each project will conclude with a comprehensive roadmap, detailing steps for real-world implementation. These roadmaps will include timelines, resource requirements, and scalable solutions, ensuring each innovation can be seamlessly integrated into Iqaluit's municipal systems and readily

adapted by other northern communities.

- **Capacity-Building Resources and Community Workshops:** To support long-term impact, the initiative will produce capacity-building resources—such as training guides, best practices, and implementation workshops. In collaboration with the University of Waterloo, we will host workshops that bring together Iqaluit officials, local stakeholders, and representatives from other Nunavut communities to explore how AI/ML and Large Language Models (LLMs) can be effectively utilized in northern municipal settings. These workshops will foster cross-community knowledge sharing, enabling broader application and adaptation of the tools developed through this partnership.

Overall, Year 3 marks a transformative phase, delivering a robust portfolio of innovations and strategies that are practical, implementable, and aligned with Iqaluit's and Nunavut's unique challenges and opportunities. The successful completion of these projects will establish a strong foundation for ongoing collaboration and future initiatives that continue to advance sustainable development across the North.

A. Need:

- Describe the need for the project.
- Explain which sector of the economy the project supports.
- Explain why this sector needs this support.
- Identify gaps that you believe the project will address.

See EOI Guide for more details and examples.

A. Need for the Project

Description of Need:
The City of Iqaluit is addressing critical gaps in municipal infrastructure management, essential service delivery, and urban planning to meet the growing Economic development demands of its residents while navigating the unique challenges of a northern, remote environment. Efficient municipal operations, particularly for essential services like water and sewer delivery, asset management, and sustainable land use planning, are vital to fostering resilience and long term sustainable growth. Incorporating advanced AI/ML technologies into these areas presents a unique opportunity for Iqaluit to improve operational efficiency, reduce environmental impact, and optimize planning decisions tailored to the city's specific context.

Supported Sector of the Economy:
The project supports economic development and growth in the City of Iqaluit, particularly through enhancing municipal governance, public infrastructure, and urban planning.

Why This Sector Needs Support:
Economic growth in Iqaluit is hindered by unique logistical, environmental, and economic challenges associated with its northern and remote location. Limited resources, high transportation costs, harsh environmental conditions, and a growing population strain the city's ability to deliver essential services and develop infrastructure effectively. Addressing these challenges is critical to creating a foundation for sustainable economic development.

By integrating advanced AI/ML technologies, this initiative enables Iqaluit to optimize resource allocation, improve service efficiency, and make data-driven decisions. These advancements support the growth of local businesses, enhance workforce capacity, and attract investment in innovative solutions tailored to the city's needs. Ultimately, these improvements strengthen Iqaluit's infrastructure and governance, providing a stable platform for long-term economic development and resilience

Identified Gaps the Project Will Address:

1. **Asset Management and Tasking (IQA-02**): Currently, Iqaluit's asset management lacks a unified, efficient system. This project will analyze existing software, identify inefficiencies, and recommend an integrated software stack to streamline municipal asset management and tasking processes. This will improve operational efficiency, reduce redundancies, and ensure effective resource allocation.

2. **Trucked Water and Sewer Services Optimization (IQA-03):** Developing predictive algorithms to optimize routing and scheduling will reduce fuel usage, minimize operational costs, and enhance service reliability for residents, addressing both financial and environmental concerns. Additionally, this approach will support economic growth by enabling the expansion of trucked water and sewer services, ensuring that development is not constrained by the limited capacity of utilidor systems. By improving efficiency and scalability, the project will empower Iqaluit to meet the demands of a growing population and facilitate sustainable urban development.

3**. AI/ML-Driven Land Use Planning (IQA-04):** Iqaluit's land use planning must account for unique factors such as permafrost, zoning requirements, and limited development space. The current time required to review these factors impacts the approval of development permits and land use plans, which can slow economic development in the community. This project will develop AI/ML tools to streamline the analysis of planning data and create optimized land use plans, enabling faster and more efficient reviews. These tools will support sustainable growth by helping the city make informed decisions that balance community needs with environmental constraints while accelerating the development process to foster economic growth.

4. **Capacity Building for Sustainable Urban Planning:** This initiative also provides a platform for knowledge transfer and capacity building by engaging Iqaluit's municipal officials with leading experts and researchers in AI, urban planning, and infrastructure management. It will create resources like training guides and workshops, fostering a culture of data-driven, sustainable development and building skills for the long term.

By addressing these gaps, this project will enhance the City of Iqaluit's operational capabilities, support sustainable development, and serve as a model for how northern municipalities can leverage AI to improve efficiency and resilience. This initiative will also engage academics, planners, and community members in a collaborative, cross-disciplinary approach, ensuring that solutions are evidence-based and relevant to the city's day-to-day needs.

Rationale for CanNor Investment

Investing in this project aligns directly with CanNor's mandate to foster sustainable economic growth, community resilience, and innovation in Canada's northern communities and specifically to look at projects that fill the AI skills gap and acquiring high-cost talent . By supporting the collaboration between the City of Iqaluit and the University of Waterloo, CanNor can play a pivotal role in demonstrating how northern communities can leverage advanced AI and ML technologies to address municipal challenges and improve essential services. The project focuses on training local representatives, ensuring capacity-building and a lasting impact on the community. Funding from CanNor would accelerate the deployment of AI-driven solutions and provide a replicable model for sustainable development across Nunavut, directly contributing to northern innovation and capacity.

Lack of Comparable Services in Nunavut and Iqaluit

Currently, there are no comparable AI and ML-driven municipal services available in Nunavut or specifically within the City of Iqaluit. This project is pioneering the integration of AI solutions for municipal management, a domain that has historically relied on conventional and labor-intensive processes that lack data optimization. The absence of similar initiatives in the North presents an opportunity for Iqaluit to set a precedent for AI-driven infrastructure management, benefiting not only local residents but also serving as a replicable framework for other communities in Nunavut. The project will provide the city with tools uniquely adapted to northern needs, such as cold-climate resilience and sustainable growth, positioning Iqaluit as a leader in advanced municipal services in the Arctic.

B. Outcomes:

- Describe what will be accomplished or produced at the end of the project.
- Explain why these outcomes are important to your business or organization.

See EOI Guide for more details and examples.

Outcomes:

At the end of this multi-year project, the collaboration between the City of Iqaluit and the University of Waterloo will produce a comprehensive suite of AI-driven solutions tailored specifically to the unique municipal needs of Iqaluit and the northern environment. Key outcomes will include:

1. **Finalized AI Tools and Solutions:** These tools will include optimized software for municipal applicability. Each tool will enable data-informed decision-making, significantly enhancing operational efficiency and supporting sustainable urban growth in Iqaluit.

2. **Implementation Roadmaps:** Detailed roadmaps for integrating these innovations into Iqaluit's municipal systems will be developed. These roadmaps will outline steps for real-world implementation, including timelines and resource requirements, ensuring these solutions are readily adaptable and scalable to meet evolving community needs.

3. **Capacity-Building Resources and Community Workshops:** To ensure lasting impact, training guides, best practices, and community workshops will be created. These resources will enable Iqaluit representatives and local stakeholders to fully utilize the new AI tools, empowering them with knowledge and skills for sustained success. The workshops will also support cross-community knowledge sharing, fostering further application of these tools in other northern communities.

4. **Cross-Community Knowledge Network**: By connecting over 200 AI-focused researchers through the University of Waterloo's network, this project will establish a knowledge-sharing platform that supports ongoing learning and innovation in northern infrastructure management.

5. **Final Project Report with Documentation and Recommendations**: A concluding report that details the development process, key insights, challenges encountered, and future recommendations to improve and scale AI/ML for other communities with similar needs.

Importance of These Outcomes

These outcomes are crucial for the City of Iqaluit and Nunavut at large, as they lay the foundation for a more resilient, data-driven approach to municipal management. By adopting AI-powered solutions, Iqaluit can address critical challenges, such as resource constraints, cold-climate logistics, and rapid urbanization. The enhanced efficiency in infrastructure management and predictive scheduling will reduce operational costs and environmental impact, which are essential for sustainable development in remote areas.

Moreover, these outcomes align with the Future Cities Institute's mission to equip communities with advanced research and tools that support sustainable municipal futures. For the City of Iqaluit, this project not only improves local service delivery but also positions the city as an innovative leader in AI-driven municipal solutions, creating a model for other northern communities to follow. Ultimately, this project will drive meaningful improvements in community resilience, economic development, and quality of life for residents in Iqaluit and beyond.

C. Benefits:

- Describe the expected measurable economic benefits of the project.
- Identify benefits for your business/organization and for the broader economy (these could include your sector, community and/or territory).
- Identify additional other benefits of your project.

See EOI Guide for more details and examples.

Expected Measurable Economic Benefits

This project will deliver measurable economic benefits in several areas critical to Iqaluit's municipal operations and the broader economy of Nunavut. These include:

Benefits for the City of Iqaluit

1. **Reduced Dependency on External Consultants**: By equipping Iqaluit and other Nunavut communities with AI tools and training, the project enhances local capacity for autonomous operation. This shift to self-sufficiency means communities can manage and optimize their infrastructure without as much reliance on costly external consultants, retaining more economic value within the community.

2. **Increased Productivity:** By automating complex municipal tasks (like route optimization and data analysis for infrastructure management, as examples), the project will free up time and resources that can be redirected to other high-priority municipal needs, leading to higher overall productivity and more effective service delivery for Iqaluit residents.

3. **Enhanced Data-Driven Decision-Making**: The AI tools will provide reliable, real-time data, empowering municipalities to make strategic, evidence-based decisions in areas like resource management, employment, and infrastructure. This informed approach will enhance the effectiveness and impact of development plans on community well-being.

4. **Enhanced Workforce Capacity:** By building capacity through training programs and hands-on experience with AI tools, Iqaluit will cultivate a skilled local workforce better equipped to manage and sustain municipal operations. This will reduce dependency on external resources and create local employment opportunities in tech-focused municipal roles.

Benefits for the Broader Northern Economy

1. **Scalability and Replication for Broader Impact**: The project is designed as a scalable model that can be adapted to other remote and Indigenous communities across Canada. By establishing a replicable framework for AI-ML integration in a northern municipal context, the project lays the foundation for widespread application and further economic development in similar contexts.

2. **Advancement in Northern AI and Technology Sectors**: For the Future Cities Institute, this project advances its mandate of addressing real-world municipal challenges through technology. It strengthens FCI's expertise in AI applications for municipal operations and showcases the University of Waterloo as a leader in AI solutions for cold-climate environments. This will open doors for future research partnerships, funding opportunities, and student enrollment in related programs.

3. **Job Creation and Skill Development:** The project will create high-skill job opportunities in both research and municipal settings. The engagement of University of Waterloo students and AI researchers will lead to the development of a talent pipeline, contributing skilled workers in Nunavut.

Additional Benefits

1. **Positioning Iqaluit as an AI- Municipal Innovation Leader**: The project establishes Iqaluit as a pioneering model for how remote northern municipalities can successfully adopt AI-driven solutions. This demonstration of innovation and resilience will inspire and serve as a blueprint for other Indigenous and remote communities across Canada, amplifying its impact across the North. We intend to showcase these municipal solutions at key national municipal events, such as the Federation of Canadian Municipalities conference. This project showcases how municipalities can innovate despite geographic and resource constraints, establishing Iqaluit as a leader in sustainable and resilient urban management in Canada's North.

D. Prior work: Explain and/or attach any work that has been completed in preparation for this project (e.g. business cases, feasibility studies, design studies). **See EOI Guide for more details and examples.**

University of Waterloo
The University of Waterloo's Future Cities Institute (FCI) collaborates with a diverse range of partners, from large multinational corporations and municipalities to smaller Indigenous communities and hamlets, focusing on critical areas such as housing, mobility, sustainable infrastructure, and the integration of artificial intelligence within municipal and hamlet settings. Through operational, applied research, FCI leverages its expertise to address complex, real-world challenges defined by city or community representatives, and other partners. By engaging students in interdisciplinary capstone projects, FCI fosters hands-on learning that directly benefits its partners. Current projects, in collaboration with municipalities and community partners, showcase a commitment to impactful research and development, enhancing sustainable growth and resilience through practical, innovative solutions. This approach not only drives forward-thinking advancements but also ensures that students and partners work together to co-create solutions that are meaningful and effective within the unique context of each community.

City of Iqaluit

The relationship between the City of Iqaluit and the University of Waterloo is grounded in a collaborative effort to address municipal infrastructure, management, growth, and technology challenges specific to Arctic conditions. This partnership has led to several foundational projects, each contributing to the broader aim of building sustainable and efficient systems tailored to Iqaluit's needs.

1. **Software Stack Analysis and Recommendation for Municipal Asset Management and Works Tasking (early development)**
Through this initiative, the University of Waterloo team has worked with Iqaluit to analyze and recommend software solutions that improve municipal asset management and optimize tasking for public works. The goal is to implement a system that streamlines operations, enhances data management, and increases efficiency across municipal departments.

2. **AI/ML-Powered Trucked Water and Sewer Service Predictive Scheduling (early development)**
Given the challenges of delivering water and sewer services in remote regions, this project uses AI/ML technology to develop predictive scheduling for trucked water and sewer services. This approach aims to optimize delivery schedules, reduce operational costs, and ensure timely service for Iqaluit residents to support economic growth by enabling the expansion of trucked water and sewer services, ensuring that development is not constrained by the limited capacity of utilidor systems.

3. **AI/ML-Powered Land Use Planning (early development)**
This project leverages AI/ML to create a data-driven approach to land use planning that directly supports economic development in Iqaluit. By analyzing current and projected data, the project aims to develop tools that facilitate sustainable and efficient land development, enabling the city to unlock new economic opportunities. These tools will address the unique spatial and environmental constraints of Arctic planning while accelerating

development timelines, attracting investment, and fostering a resilient and growth-oriented local economy.

4. **AI/ML Finance ERP Integration for Budgeting (proposed)**
This project will integrate AI/ML into Iqaluit's financial ERP system to automate budgeting, streamline financial processes, and improve fiscal forecasting. By enhancing efficiency and enabling data-driven decision-making, it supports economic development by optimizing resource allocation and fostering financial sustainability.

5.**AI/ML-Supported HR Onboarding and Training (proposed)**
This project will use AI/ML tools to streamline onboarding and training for municipal staff, tailoring processes to individual roles and improving skill development. By building a more skilled and efficient workforce, it supports economic growth by enhancing service delivery and workforce retention.

Through these projects, the University of Waterloo and the City of Iqaluit have established a strong foundation for impactful municipal projects that are scalable to other Arctic communities. This groundwork, including staff training, pilot initiatives, and technology assessment, enables Iqaluit to serve as a model for other Nunavut communities seeking tailored infrastructure and asset management solutions.

E. Partners/supporters: List partners and/or supporters in the project (including their involvement). **See EOI Guide for more details and examples.**

1. **University of Waterloo Future Cities Institute (FCI):** The Future Cities Institute at UWaterloo, founded by CAIVAN, will support this project by integrating resources from a global network of interdisciplinary researchers, industry leaders, students, and community members focused on sustainable municipal development. FCI will provide access to UWaterloo's advanced computing resources with AI/ML capabilities, specialized software for AI and language model development, and powerful data analysis tools. Ranked as Canada's #1 comprehensive research university (Research Information 2023), UWaterloo also brings expertise from Canada Excellence Research Chairs in Human-Centred Robotics, Machine Intelligence, Developmental Science, and Data Intelligence to advance project goals.

2. **Waterloo Data and Artificial Intelligence Institute (Waterloo.AI):** Waterloo.AI offers participants intensive, short courses (3-4 days) led by top instructors in AI, allowing project team members to deepen their technical expertise. These courses are tailored to bridge academic AI knowledge with practical applications, providing participants the skills to integrate new AI techniques directly into industry contexts.

3. **UWaterloo Faculty of Engineering:** As the largest engineering faculty in Canada, UWaterloo's Faculty of Engineering supports the project by providing resources, expertise, and tools required for AI-driven urban solutions. The faculty fosters opportunities for students to engage with AI through applied design projects, educational workshops, and hackathons, directly contributing to innovative project outcomes.

4. **UWaterloo Faculty of Mathematics:** The Faculty of Mathematics, home to the largest Computer Science department in Canada, offers additional resources and support for complex computational projects. With advanced capabilities in data science and AI, the faculty is integral in applying rigorous data intelligence approaches to urban challenges.

Forecasted Project Costs and Funding Sources						
Forecasted total project cost ($) $667,000			Amount requested from CanNor ($) $400,000			
Forecasted Budget and Anticipated Sources of Funds						
Funding Partners	Total budget	Year 1	Year 2	Year 3	Year 4	Year 5
CanNor	$400,000	$125,000	$130,000	$145,000		
Applicant	$60,000	$20,000	$20,000	$20,000		
Future Cities Institute at the University of Waterloo	$207,000	$65,000	$69,000	$73,000		

Additional funders (please specify)						
Additional funders (please specify)						
Additional funders (please specify)						
Total						

Have you already incurred costs or made legal commitments to the project? ○ Yes ● No

If yes, please describe

Diversity and Inclusion

The Government of Canada recognizes that many under-represented groups face unique economic challenges. Understanding that participation of underrepresented groups is an integral part of building strong and inclusive communities, and helps the Government of Canada provide inclusive programming to support all groups.

Priority may be given to applications that can demonstrate how they support inclusive growth.

Is your business owned or majority-owned or is your organization led or majority-led by one or more of the following under-represented groups (check all that apply)?

- ☐ Women
- ☐ Indigenous Peoples (First Nations)
- ☐ Visible Minorities
- X Indigenous Peoples (Inuit)
- ☐ Indigenous Peoples (Metis)
- ☐ Members of Official French Language Minority Communities
- ☐ Youth (30 years or less)
- ☐ Persons with Disabilities
- ☐ Newcomers to Canada
- ☐ Black Communities
- ☐ Other Racialized Communities
- ☐ 2SLGBTQI+Others (list):

If applicable, please indicate if your project includes or supports one or more of the following underrepresented groups:

- X Women
- ☐ Indigenous Peoples (First Nations)
- ☐ Visible Minorities
- X Indigenous Peoples (Inuit)
- ☐ Indigenous Peoples (Metis)
- ☐ Members of Official French Language Minority Communities
- X Youth (30 years or less)
- ☐ Persons with Disabilities
- ☐ Newcomers to Canada
- ☐ Black Communities

☐ Other Racialized Communities ☐ 2SLGBTQI+ ☐ Others (list):

Authorization		
This section is to be completed by a representative of the applicant duly authorized to make this declaration in submission of a funding application to the Canadian Northern Economic Development Agency.		
Representative's legal name Bill Williams	Title Senior Executive Director of Municipal Infrastructure and Planning	Date (YYYYMMDD)

Project Submission and CanNor Contact Information
To submit the application and for general inquiries, send an email to: operations@cannor.gc.ca In the subject line, use the following format to assist CanNor in processing your application. **Email Subject Line Format:** 2025-2026 EOI – [Territory (i.e. NU, NWT or YT)] – [Organization Name] ***Example**: 2025-2026 EOI – NU – ABC Enterprises*

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